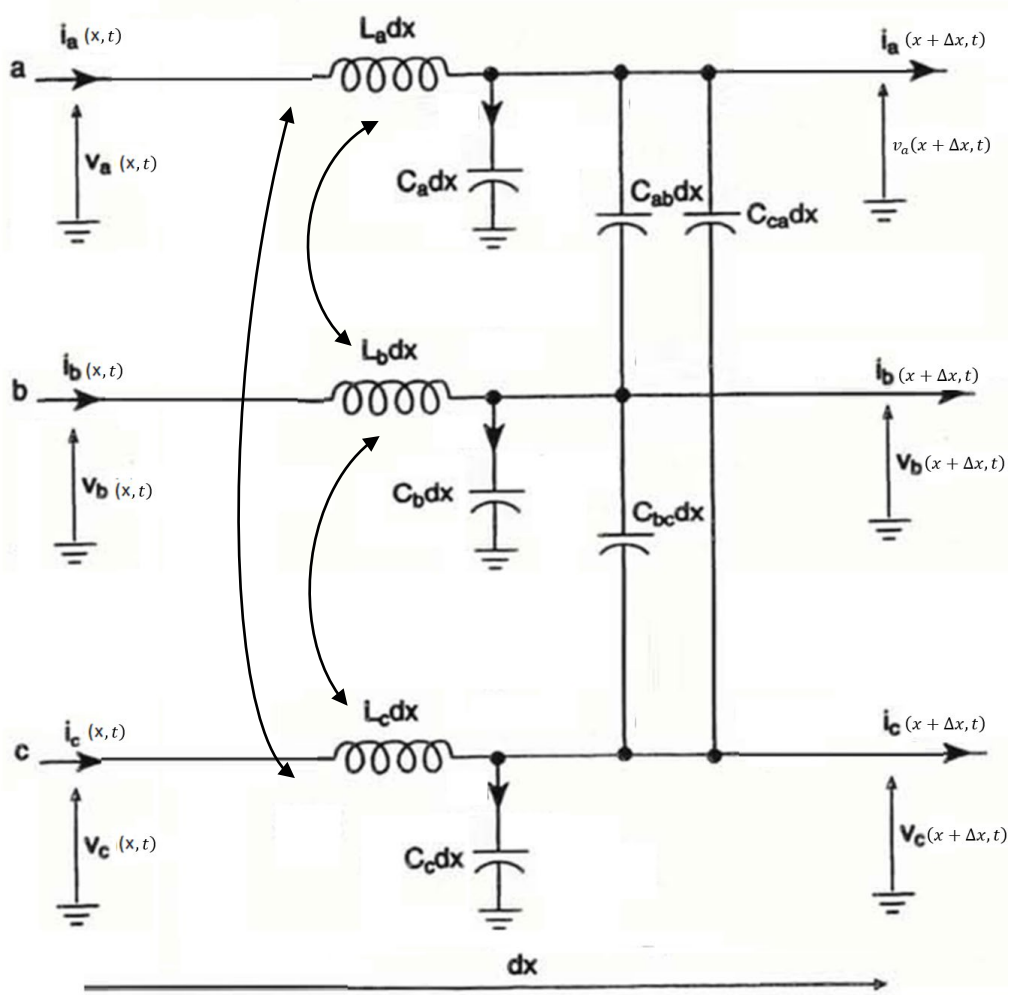
# Power System Protection II

**EE 511**

The materials of this set of slides can be found in chapter 13 of the following book.
Duncan Glover, Mulukutla S. Sarma, Thomas Overbye, “Power Systems Analysis and Design,” CL-Engineering; 5 edition

# MULTICONDUCTOR LINES



**For the 'a' phase:**

$$\Delta v_a = v_a(x+\Delta x,t) - v_a(x,t) = -(L_a \Delta x \frac{\partial i_a(x,t)}{\partial t} + L_{ab}\Delta x \frac{\partial i_b(x,t)}{\partial t} + L_{ac}\Delta x \frac{\partial i_c(x,t)}{\partial t})$$

Therefor,

$$\frac{v_a(x+\Delta x,t) - v_a(x,t)}{\Delta x} = \frac{\partial v_a(x,t)}{\partial x} = -(L_a \frac{\partial i_a(x,t)}{\partial t} + L_{ab} \frac{\partial i_b(x,t)}{\partial t} + L_{ac} \frac{\partial i_c(x,t)}{\partial t})$$

Similarly

$$\frac{\partial v_b(x,t)}{\partial x} = -(L_{ba} \frac{\partial i_a(x,t)}{\partial t} + L_b \frac{\partial i_b(x,t)}{\partial t} + L_{bc} \frac{\partial i_c(x,t)}{\partial t})$$

$$\frac{\partial v_c(x,t)}{\partial x} = -(L_{ca} \frac{\partial i_a(x,t)}{\partial t} + L_{cb} \frac{\partial i_b(x,t)}{\partial t} + L_c \frac{\partial i_c(x,t)}{\partial t})$$

$$v_a(x+\Delta x,t) = v_a(x,t) - (L_a\Delta x\frac{\partial i_a(x,t)}{\partial t} + L_{ab}\Delta x\frac{\partial i_b(x,t)}{\partial t} + L_{ac}\Delta x\frac{\partial i_c(x,t)}{\partial t})$$

$$v_b(x+\Delta x,t) = v_b(x,t) - (L_{ba}\Delta x\frac{\partial i_a(x,t)}{\partial t} + L_b\Delta x\frac{\partial i_b(x,t)}{\partial t} + L_{bc}\Delta x\frac{\partial i_c(x,t)}{\partial t})$$

$$v_c(x+\Delta x,t) = v_c(x,t) - (L_{ca}\Delta x\frac{\partial i_a(x,t)}{\partial t} + L_{cb}\Delta x\frac{\partial i_b(x,t)}{\partial t} + L_c\Delta x\frac{\partial i_c(x,t)}{\partial t})$$

$$\Delta i_a = i_a(x+\Delta x,t) - i_a(x,t)$$

$$= -(c_a\Delta x\frac{\partial v_a(x+\Delta x,t)}{\partial t} + c_{ab}\Delta x\frac{\partial v_{ab}(x+\Delta x,t)}{\partial t} + c_{ac}\Delta x\frac{\partial v_{ac}(x+\Delta x,t)}{\partial t})$$

$$= -(c_a\Delta x\frac{\partial v_a(x+\Delta x,t)}{\partial t} + c_{ab}\Delta x\frac{\partial(v_a(x+\Delta x,t) - v_b(x+\Delta x,t))}{\partial t}$$

$$+ c_{ac}\Delta x\frac{\partial(v_a(x+\Delta x,t) - v_c(x+\Delta x,t))}{\partial t})$$

$$= -\left((c_a + c_{ab} + c_{ac})\Delta x\frac{\partial v_a(x+\Delta x,t)}{\partial t} - c_{ab}\Delta x\frac{\partial v_b(x+\Delta x,t)}{\partial t} - c_{ac}\Delta x\frac{\partial v_c(x+\Delta x,t)}{\partial t}\right)$$

$$= -\Big((c_a + c_{ab} + c_{ac})\Delta x \frac{\partial(v_a(x,t) - (L_a \Delta x \frac{\partial i_a(x,t)}{\partial t} + L_{ab}\Delta x \frac{\partial i_b(x,t)}{\partial t} + L_{ac}\Delta x \frac{\partial i_c(x,t)}{\partial t}))}{\partial t}$$

$$- c_{ab}\Delta x \frac{\partial\left(v_b(x,t) - \left(L_{ba}\Delta x \frac{\partial i_a(x,t)}{\partial t} + L_b \Delta x \frac{\partial i_b(x,t)}{\partial t} + L_{bc}\Delta x \frac{\partial i_c(x,t)}{\partial t}\right)\right)}{\partial t}$$

$$- c_{ac}\Delta x \frac{\partial(v_c(x,t) - (L_{ca}\Delta x \frac{\partial i_a(x,t)}{\partial t} + L_{cb}\Delta x \frac{\partial i_b(x,t)}{\partial t} + L_c \Delta x \frac{\partial i_c(x,t)}{\partial t}))}{\partial t}\Big)$$

$$= -\Big((c_a + c_{ab} + c_{ac})\Delta x \frac{\partial v_a(x,t)}{\partial t} - c_{ab}\Delta x \frac{\partial v_b(x,t)}{\partial t} - c_{ac}\Delta x \frac{\partial v_c(x,t)}{\partial t}\Big)$$

$$-c_a L_a \Delta x^2 \frac{\partial^2 i_a(x,t)}{\partial t^2} - c_a\, L_{ab}\Delta x^2 \frac{\partial^2 i_b(x,t)}{\partial t^2} - c_a\, L_{ac}\Delta x^2 \frac{\partial^2 i_c(x,t)}{\partial t^2}$$

$$-c_{ab} L_{ba}\Delta x^2 \frac{\partial^2 i_a(x,t)}{\partial t^2} - c_{ab}\, L_b \Delta x^2 \frac{\partial^2 i_b(x,t)}{\partial t^2} - c_{ab}\, L_{bc}\Delta x^2 \frac{\partial^2 i_c(x,t)}{\partial t^2}$$

$$-c_{ac} L_{ca}\Delta x^2 \frac{\partial^2 i_a(x,t)}{\partial t^2} - c_{ac}\, L_{cb}\Delta x^2 \frac{\partial^2 i_b(x,t)}{\partial t^2} - c_{ac}\, L_c \Delta x^2 \frac{\partial^2 i_c(x,t)}{\partial t^2}$$

Terms that include $\Delta x^2$ are ignored. Therefore,

$$\Delta i_a = i_a(x+\Delta x, t) - i_a(x,t)$$
$$= -\left((c_a + c_{ab} + c_{ac})\Delta x \frac{\partial v_a(x,t)}{\partial t} - c_{ab}\Delta x \frac{\partial v_b(x,t)}{\partial t} - c_{ac}\Delta x \frac{\partial v_c(x,t)}{\partial t}\right)$$

Therefore,

$$\frac{i_a(x+\Delta x, t) - i_a(x,t)}{\Delta x} = \frac{\partial i_a(x,t)}{\partial x}$$
$$= -\left((c_a + c_{ab} + c_{ac})\frac{\partial v_a(x,t)}{\partial t} - c_{ab}\frac{\partial v_b(x,t)}{\partial t} - c_{ac}\frac{\partial v_c(x,t)}{\partial t}\right)$$

Similarly:

$$\frac{\partial i_b(x,t)}{\partial x} = -\left(-c_{ba}\frac{\partial v_a(x,t)}{\partial t}(c_{ba} + c_b + c_{bc})\frac{\partial v_b(x,t)}{\partial t} - c_{bc}\frac{\partial v_c(x,t)}{\partial t}\right)$$
$$\frac{\partial i_c(x,t)}{\partial x} = -\left(-c_{ca}\frac{\partial v_a(x,t)}{\partial t} - c_{cb}\frac{\partial v_b(x,t)}{\partial t} + (c_{ca} + c_{cb} + c_c)\frac{\partial v_c(x,t)}{\partial t}\right)$$

In summary:

$$\frac{\partial i_a(x,t)}{\partial x} = -\left((c_a + c_{ab} + c_{ac})\frac{\partial v_a(x,t)}{\partial t} - c_{ab}\frac{\partial v_b(x,t)}{\partial t} - c_{ac}\frac{\partial v_c(x,t)}{\partial t}\right)$$

$$\frac{\partial i_b(x,t)}{\partial x} = -\left(-c_{ba}\frac{\partial v_a(x,t)}{\partial t}(c_{ba}+c_b + c_{bc})\frac{\partial v_b(x,t)}{\partial t} - c_{bc}\frac{\partial v_c(x,t)}{\partial t}\right)$$

$$\frac{\partial i_c(x,t)}{\partial x} = -\left(-c_{ca}\frac{\partial v_a(x,t)}{\partial t} - c_{cb}\frac{\partial v_b(x,t)}{\partial t} + (c_{ca} + c_{cb} + c_c)\frac{\partial v_c(x,t)}{\partial t}\right)$$

$$\frac{\partial v_a(x,t)}{\partial x} = -(L_a\frac{\partial i_a(x,t)}{\partial t} + L_{ab}\frac{\partial i_b(x,t)}{\partial t} + L_{ac}\frac{\partial i_c(x,t)}{\partial t})$$

$$\frac{\partial v_b(x,t)}{\partial x} = -(L_{ba}\frac{\partial i_a(x,t)}{\partial t} + L_b\frac{\partial i_b(x,t)}{\partial t} + L_{bc}\frac{\partial i_c(x,t)}{\partial t})$$

$$\frac{\partial v_c(x,t)}{\partial x} = -(L_{ca}\frac{\partial i_a(x,t)}{\partial t} + L_{cb}\frac{\partial i_b(x,t)}{\partial t} + L_c\frac{\partial i_c(x,t)}{\partial t})$$

Assuming the transmission line is fully transposed we have

La=Lb=Lc=Ls and Lab=Lbc=Lca=Lm

Ca=Cb=Cc and Cab=Cbc=Cca

Cs=(Ca+Cab+Cca)= (Cab+Cb+Cbc)=(Cca+Cbc+Cc)

Cm=-Cab=-Cbc=-Cca

$$\frac{\partial v(x,t)}{\partial x} = -L\frac{\partial i(x,t)}{\partial t} \qquad \frac{\partial i(x,t)}{\partial x} = -C\frac{\partial v(x,t)}{\partial t}$$

$$\mathbf{L} = \begin{bmatrix} L_s & L_m & L_m \\ L_m & L_s & L_m \\ L_m & L_m & L_s \end{bmatrix} \text{ H/m} \qquad \mathbf{C} = \begin{bmatrix} C_s & C_m & C_m \\ C_m & C_s & C_m \\ C_m & C_m & C_s \end{bmatrix} \text{ F/m}$$

$$\boldsymbol{v} = \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \qquad \boldsymbol{i} = \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

Note that $L_s$, $L_m$, and $C_s$ are positive, whereas $C_m$ is negative.

Calculating the laplace transform

$$\frac{d\mathbf{V}(x,s)}{dx} = -s\mathbf{L}\mathbf{I}(x,s) \tag{13.6.1}$$

$$\frac{d\mathbf{I}(x,s)}{dx} = -s\mathbf{C}\mathbf{V}(x,s) \tag{13.6.2}$$

where

$$\mathbf{V}(x,s) = \begin{bmatrix} \mathbf{V}_{ag}(x,s) \\ \mathbf{V}_{bg}(x,s) \\ \mathbf{V}_{cg}(x,s) \end{bmatrix} \qquad \mathbf{I}(x,s) = \begin{bmatrix} \mathbf{I}_a(x,s) \\ \mathbf{I}_b(x,s) \\ \mathbf{I}_c(x,s) \end{bmatrix} \tag{13.6.3}$$

Equations (13.6.1) and (13.6.2) are identical to (13.1.5) and (13.1.6) except that scalar quantities are replaced by vector quantities. $\mathbf{V}(x,s)$ is the vector of line-to-ground voltages and $\mathbf{I}(x,s)$ is the vector of line currents. For a completely transposed line, the $3 \times 3$ inductance matrix $\mathbf{L}$ and capacitance matrix $\mathbf{C}$ are given by

$$\mathbf{L} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}_m & \mathbf{L}_m \\ \mathbf{L}_m & \mathbf{L}_s & \mathbf{L}_m \\ \mathbf{L}_m & \mathbf{L}_m & \mathbf{L}_s \end{bmatrix} \quad \mathrm{H/m} \tag{13.6.4}$$

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_s & \mathbf{C}_m & \mathbf{C}_m \\ \mathbf{C}_m & \mathbf{C}_s & \mathbf{C}_m \\ \mathbf{C}_m & \mathbf{C}_m & \mathbf{C}_s \end{bmatrix} \quad \mathrm{F/m} \tag{13.6.5}$$

We now transform the phase quantities to modal quantities. First, we define

$$\begin{bmatrix} \mathbf{V}_{ag}(x,s) \\ \mathbf{V}_{bg}(x,s) \\ \mathbf{V}_{cg}(x,s) \end{bmatrix} = \mathbf{T}_\mathbf{V} \begin{bmatrix} \mathbf{V}^0(x,s) \\ \mathbf{V}^+(x,s) \\ \mathbf{V}^-(x,s) \end{bmatrix} \tag{13.6.6}$$

$$\begin{bmatrix} \mathbf{I}_a(x,s) \\ \mathbf{I}_b(x,s) \\ \mathbf{I}_c(x,s) \end{bmatrix} = \mathbf{T}_\mathbf{I} \begin{bmatrix} \mathbf{I}^0(x,s) \\ \mathbf{I}^+(x,s) \\ \mathbf{I}^-(x,s) \end{bmatrix} \tag{13.6.7}$$

$\mathbf{V}^0(x,s)$, $\mathbf{V}^+(x,s)$, and $\mathbf{V}^-(x,s)$ are denoted *zero-mode*, *positive-mode*, and *negative-mode* voltages, respectively. Similarly, $\mathbf{I}^0(x,s)$, $\mathbf{I}^+(x,s)$, and $\mathbf{I}^-(x,s)$ are *zero-*, *positive-*, and *negative-mode* currents. $\mathbf{T}_\mathbf{V}$ and $\mathbf{T}_\mathbf{I}$ are $3 \times 3$ constant transformation matrices, soon to be specified. Denoting $\mathbf{V}_m(x,s)$ and $\mathbf{I}_m(x,s)$ as the modal voltage and modal current vectors,

$$\mathbf{V}(x,s) = \mathbf{T}_\mathbf{V}\mathbf{V}_m(x,s) \tag{13.6.8}$$

$$\mathbf{I}(x,s) = \mathbf{T}_\mathbf{I}\mathbf{I}_m(x,s) \tag{13.6.9}$$

Using (13.6.8) and (13.6.9) in (13.6.1),

$$\mathbf{T}_\mathbf{V} \frac{d\mathbf{V}_m(x,s)}{dx} = -s\mathbf{L}\mathbf{T}_\mathbf{I}\mathbf{I}_m(x,s)$$

or

$$\frac{d\mathbf{V}_m(x,s)}{dx} = -s(\mathbf{T}_v^{-1}\mathbf{L}\mathbf{T}_\mathrm{I})\mathbf{I}_m(x,s) \tag{13.6.10}$$

Similarly, using (13.6.8) and (13.6.9) in (13.6.2),

$$\frac{d\mathbf{I}_m(x,s)}{dx} = -s(\mathbf{T}_\mathrm{I}^{-1}\mathbf{C}\mathbf{T}_v)\mathbf{V}_m(x,s) \tag{13.6.11}$$

The objective of the modal transformation is to diagonalize the matrix products within the parentheses of (13.6.10) and (13.6.11), thereby decoupling these vector equations. For a three-phase completely transposed line, $\mathbf{T}_\mathrm{V}$ and $\mathbf{T}_\mathrm{I}$ are given by

$$\mathbf{T}_\mathrm{V} = \mathbf{T}_\mathrm{I} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \tag{13.6.12}$$

Also, the inverse transformation matrices are

$$\mathbf{T}_\mathrm{V}^{-1} = \mathbf{T}_\mathrm{I}^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \tag{13.6.13}$$

Substituting (13.6.12), (13.6.13), (13.6.4), and (13.6.5) into (13.6.10) and (13.6.11) yields

$$\frac{d}{dx}\begin{bmatrix} \mathrm{V}^0(x,s) \\ \mathrm{V}^+(x,s) \\ \mathrm{V}^-(x,s) \end{bmatrix} = \begin{bmatrix} -s(\mathrm{L}_s + 2\mathrm{L}_m) & 0 & 0 \\ 0 & -s(\mathrm{L}_s - \mathrm{L}_m) & 0 \\ 0 & 0 & -s(\mathrm{L}_s - \mathrm{L}_m) \end{bmatrix} \times \begin{bmatrix} \mathrm{I}^0(x,s) \\ \mathrm{I}^+(x,s) \\ \mathrm{I}^-(x,s) \end{bmatrix} \tag{13.6.14}$$

$$\frac{d}{dx}\begin{bmatrix} \mathrm{I}^0(x,s) \\ \mathrm{I}^+(x,s) \\ \mathrm{I}^-(x,s) \end{bmatrix} = \begin{bmatrix} -s(\mathrm{C}_s + 2\mathrm{C}_m) & 0 & 0 \\ 0 & -s(\mathrm{C}_s - \mathrm{C}_m) & 0 \\ 0 & 0 & -s(\mathrm{C}_s - \mathrm{C}_m) \end{bmatrix} \times \begin{bmatrix} \mathrm{V}^0(x,s) \\ \mathrm{V}^+(x,s) \\ \mathrm{V}^-(x,s) \end{bmatrix} \tag{13.6.15}$$

From (13.6.14) and (13.6.15), the zero-mode equations are

$$\frac{d\mathrm{V}^0(x,s)}{dx} = -s(\mathrm{L}_s + 2\mathrm{L}_m)\mathrm{I}^0(x,s) \tag{13.6.16}$$

$$\frac{d\mathrm{I}^0(x,s)}{dx} = -s(\mathrm{C}_s + 2\mathrm{C}_m)\mathrm{V}^0(x,s) \tag{13.6.17}$$

These equations are identical in form to those of a two-wire lossless line, (13.1.5) and (13.1.6). By analogy, the zero-mode waves travel at velocity

$$v^0 = \frac{1}{\sqrt{(L_s + 2L_m)(C_s + 2C_m)}} \quad \text{m/s} \tag{13.6.18}$$

And the zero-mode characteristic impedance (also called surge impedance) is

$$Z_c^0 = \sqrt{\frac{L_s + 2L_m}{C_s + 2C_m}} \quad \Omega \tag{13.6.19}$$

Similarly, the positive- and negative-mode velocities and surge impedances are

$$v^+ = v^- = \frac{1}{\sqrt{(L_s - L_m)(C_s - C_m)}} \quad \text{m/s} \tag{13.6.20}$$

$$Z_c^+ = Z_c^- = \sqrt{\frac{L_s - L_m}{C_s - C_m}} \quad \Omega \tag{13.6.21}$$